

Zeros, Multiplicity, and Graph Behavior

Answer Key

Algebra 2 Polynomial Factoring, Roots, and Graphs

Quick Answers

1. $x(x-2)(x+2)$

2. $-1, 3$

3. $x(x-5)(x+2)$

4. $-\infty, \infty$

5. $0, 4$

6. $x^3 - 3x + 2$

7. $-2, -1, 1, 2$

8. ∞, ∞

9. $x^3 - x^2 - 8x + 12$

10. $-2, 0, 3$

1. Factor $f(x) = x^3 - 4x$ and list its zeros.

Pull the greatest common factor first — skipping it is what loses a zero: $x^3 - 4x = x(x^2 - 4)$.

The second factor is a difference of squares, $x^2 - 4 = (x - 2)(x + 2)$. $x(x - 2)(x + 2)$

Each factor is linear and appears once, so the zeros are $x = 0$, $x = 2$, $x = -2$, each of multiplicity 1: the graph crosses the axis at all three.

2. $g(x) = (x + 1)(x - 3)^2$: zeros, multiplicity, and behavior.

The polynomial is already factored, so read the zeros straight off: setting each factor to zero gives $x = -1$ from $(x + 1)$ and $x = 3$ from $(x - 3)^2$.

$x = -1$ has multiplicity 1 (odd) — the graph *crosses* there. $x = 3$ has multiplicity 2 (even) — the graph *touches* the axis and turns back, without changing sign. $x = -1, x = 3$

Degree check: $1 + 2 = 3$, so g is cubic with a positive leading coefficient.

3. Factor $h(x) = x^3 - 3x^2 - 10x$ and list its zeros.

GCF first: $x^3 - 3x^2 - 10x = x(x^2 - 3x - 10)$. Then factor the trinomial — two numbers multiplying to -10 and adding to -3 are -5 and 2 : $x^2 - 3x - 10 = (x - 5)(x + 2)$.

$x(x - 5)(x + 2)$

Zeros $x = 0$, $x = 5$, $x = -2$, all of multiplicity 1, so the graph crosses at each.

4. End behavior of $f(x) = -2x^3 + 5x - 1$.

Only the leading term matters as $|x|$ grows, because $-2x^3$ outgrows $5x$ and the constant. The degree is 3 (odd), so the two ends go opposite ways, and the leading coefficient -2 is negative, so the right end falls.

$$\text{as } x \rightarrow \infty, f(x) \rightarrow \boxed{-\infty} \qquad \text{as } x \rightarrow -\infty, f(x) \rightarrow \boxed{+\infty}$$

Expected sentence: “The odd degree makes the ends opposite; the negative leading coefficient decides which end goes down.”

5. $p(x) = x^2(x - 4)^3$: degree, zeros, multiplicity, behavior.

Adding the exponents gives the degree: $2 + 3 = 5$, so p is quintic with a positive leading coefficient.

$x = 0$ has multiplicity 2 (even) — the graph *touches* the axis and turns back. $x = 4$ has multiplicity 3 (odd and greater than one) — the graph *crosses*, but flattens against the axis as it passes.

$$x = 0, x = 4$$

6. Least-degree polynomial crossing at $x = -2$ and bouncing at $x = 1$.

Crossing means an odd multiplicity, and the least odd multiplicity is 1, so $(x + 2)$ appears once. Bouncing means an even multiplicity, least is 2, so $(x - 1)^2$. With leading coefficient 1:

$$(x + 2)(x - 1)^2 = (x + 2)(x^2 - 2x + 1) = x^3 - 2x^2 + x + 2x^2 - 4x + 2$$

Collecting like terms gives the expanded form.

$$x^3 - 3x + 2$$

Check: substituting $x = 1$ gives $1 - 3 + 2 = 0$ ✓ and $x = -2$ gives $-8 + 6 + 2 = 0$ ✓

7. Solve $x^4 - 5x^2 + 4 = 0$.

Treat it as a quadratic in x^2 : two numbers multiplying to 4 and adding to -5 are -1 and -4 , so $x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$. Each factor is a difference of squares:

$$(x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2)$$

Four distinct linear factors, each of multiplicity 1, so the graph crosses at all four.

$$x = -2, -1, 1, 2$$

8. End behavior of $g(x) = 3x^4 - x^3$.

The leading term is $3x^4$. The degree is 4 (even), so both ends travel the same way, and the leading coefficient 3 is positive, so both rise.

$$\text{as } x \rightarrow \infty, g(x) \rightarrow \boxed{+\infty} \qquad \text{as } x \rightarrow -\infty, g(x) \rightarrow \boxed{+\infty}$$

Expected sentence: “Problem 4 had odd degree, so its ends split; here the degree is even, so a negative input raised to an even power is still positive and both ends rise.”

9. Least-degree polynomial crossing at $x = -3$ and touching at $x = 2$.

Crossing at $x = -3$ gives the factor $(x + 3)$ once; touching at $x = 2$ gives $(x - 2)^2$. With leading coefficient 1:

$$(x + 3)(x - 2)^2 = (x + 3)(x^2 - 4x + 4) = x^3 - 4x^2 + 4x + 3x^2 - 12x + 12$$

Collecting like terms gives the expanded form.

$$x^3 - x^2 - 8x + 12$$

Check the sketch: degree 3 is odd with a positive leading coefficient, so the curve falls on the left and rises on the right, bouncing off the axis at $x = 2$ on the way up.

10. Sketch $y = x(x + 2)^2(x - 3)$.

Zeros and behavior. $x = 0$ (multiplicity 1, crosses), $x = -2$ (multiplicity 2, touches), $x = 3$ (multiplicity 1, crosses). $x = -2, x = 0, x = 3$

End behavior. Degree $1 + 2 + 1 = 4$ (even) with a positive leading coefficient, so both ends rise.

Model sketch (manual review — compare the student's curve to this description).

Coming down from the upper left, the curve touches the axis at $x = -2$ without crossing, turns back up, then comes down through $x = 0$, dips to a minimum between $x = 0$ and $x = 3$ (the low point is about $y = -32$, at $x = 2$), crosses at $x = 3$, and rises steeply. *Credit the sketch when the three intercepts are marked, the bounce at $x = -2$ is visible, and both ends point up; the vertical scale is the student's own choice.*